Midterm-Prep Recitation

The Midterm is Mostly Conceptual

1. Topics Covered:
    a. Naive Bayes
    b. K Nearest Neighbors
    c. Decision Tree
    d. Linear Regression
    e. K-means and Hierarchical Clustering
    f. Logistic Regression
    g. SVM

In general, questions where you are given some parameters and you must plug numbers into equations will not be there.

No coding questions.

These slides would be very helpful for making a cheat sheet!

Brief Recap: Naive Bayes

Assumes features are conditionally independent.

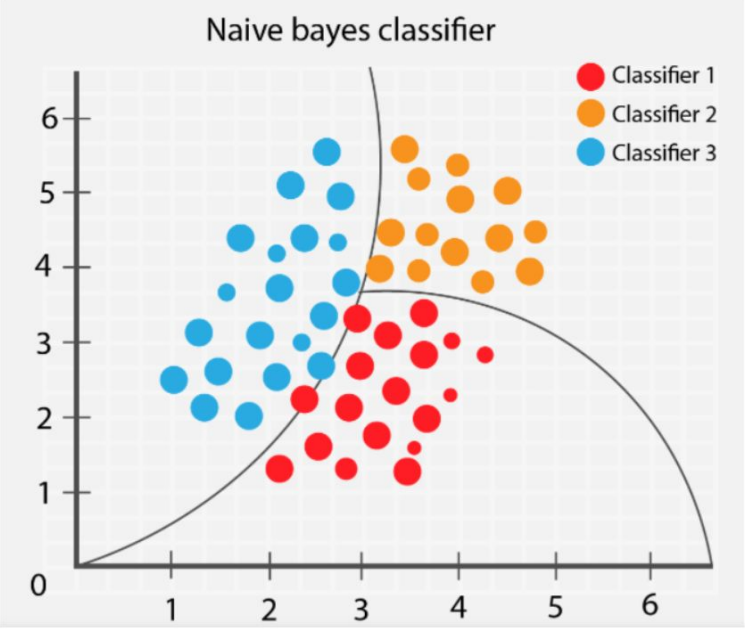
A = class, B = features

$$P(A|B) = \frac{P(B|A) * P(A)}{P(B)}$$

Supervised, Generative, Probabilistic, non-linear

Training: O(nd)

Prediction: O(d)



Brief Recap: kNN

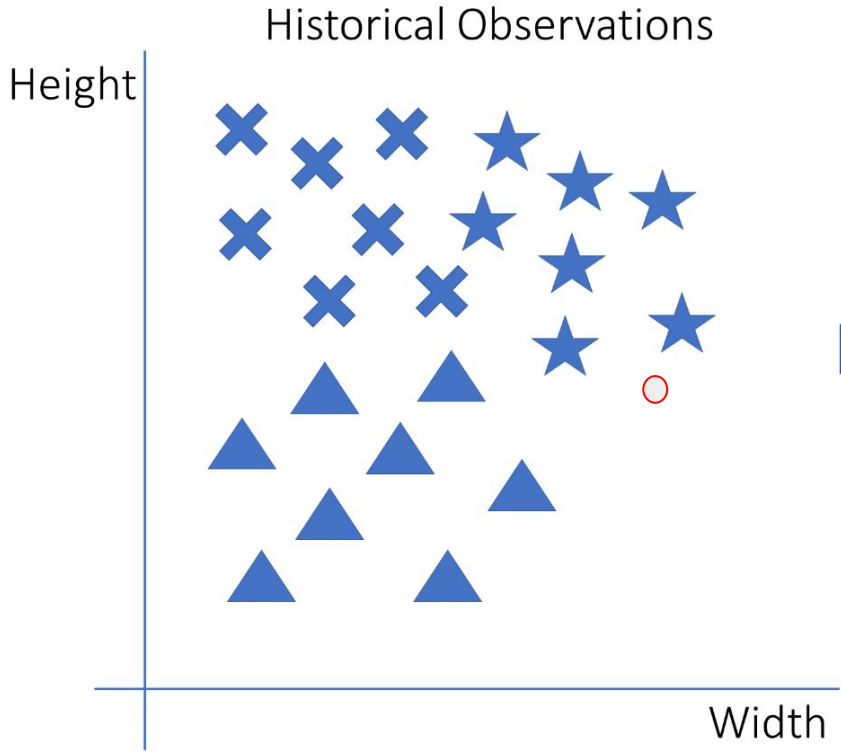
Given points with class labels, compute the distance of a new point to all points, and label the new point based on k-nearest neighbors.

Supervised, Discriminative, deterministic*, non-linear

Training: N/A

Prediction: O(nd)

- How big should k be?
- What happens at extremes of k?



Brief Recap: Decision Tree

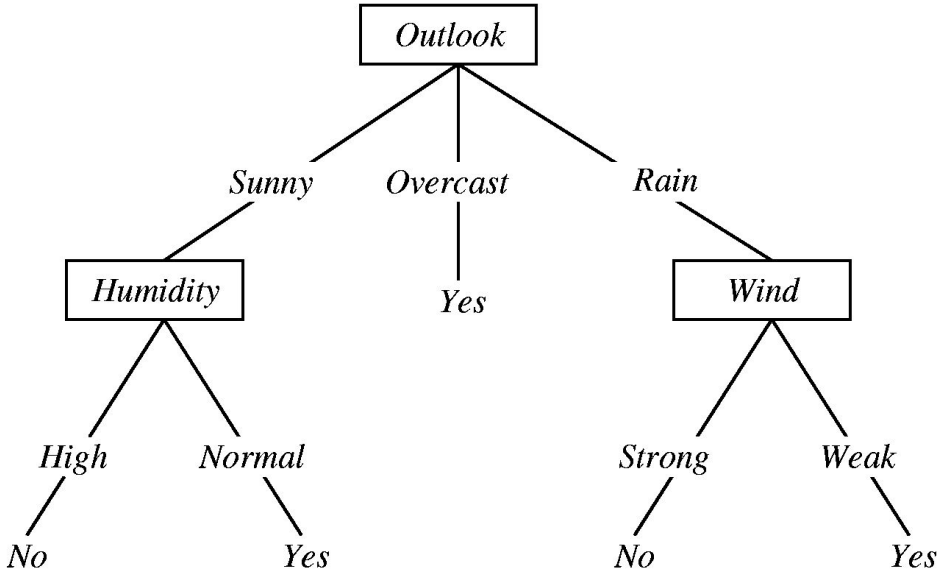
Computes maximum information gain of each split on features.

Supervised, Discriminative, probabilistic*, non-linear

Training: O(ndlog(n))

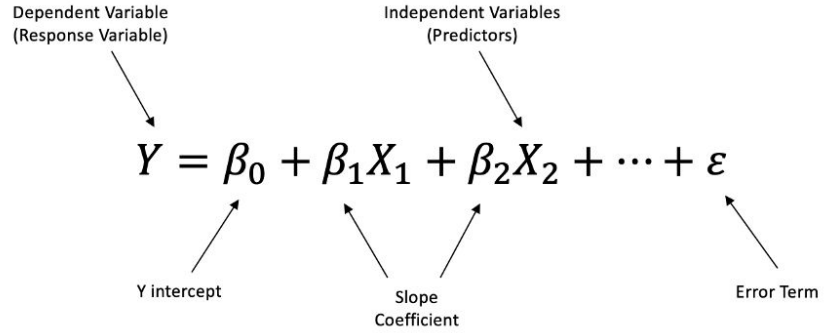
Prediction: O(n)

- Can you overfit?



Brief Recap: Linear Regression

Learn linear weights of each feature such that we find the line of best fit.

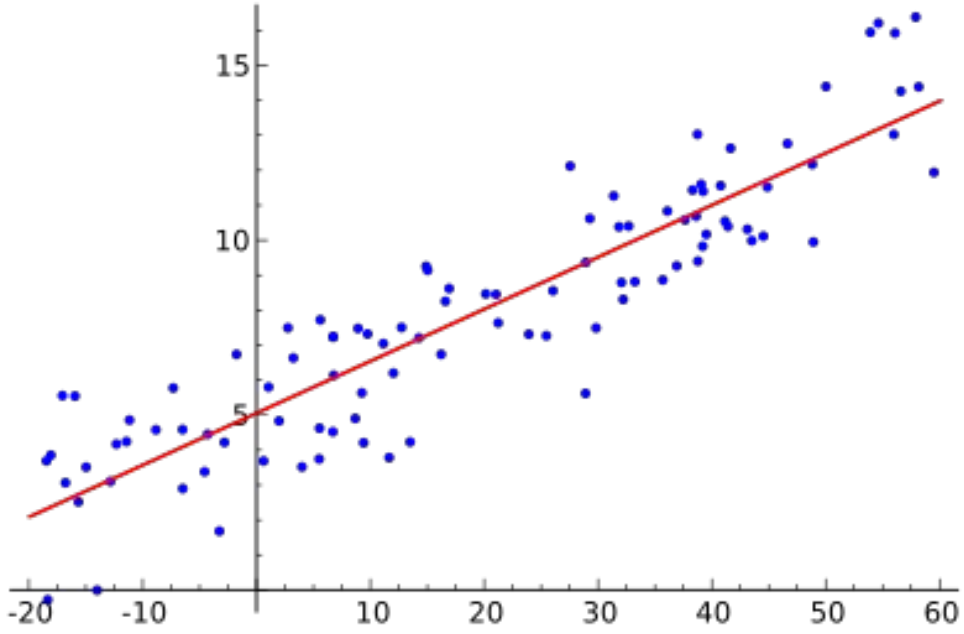


Deterministic, not a classifier

Training: O(nd^2)

Prediction: O(d)

- How to solve multivariate regression?
- Difference between multivariate and univariate?
- What if we have more features than samples?
- How to reduce amount of features?



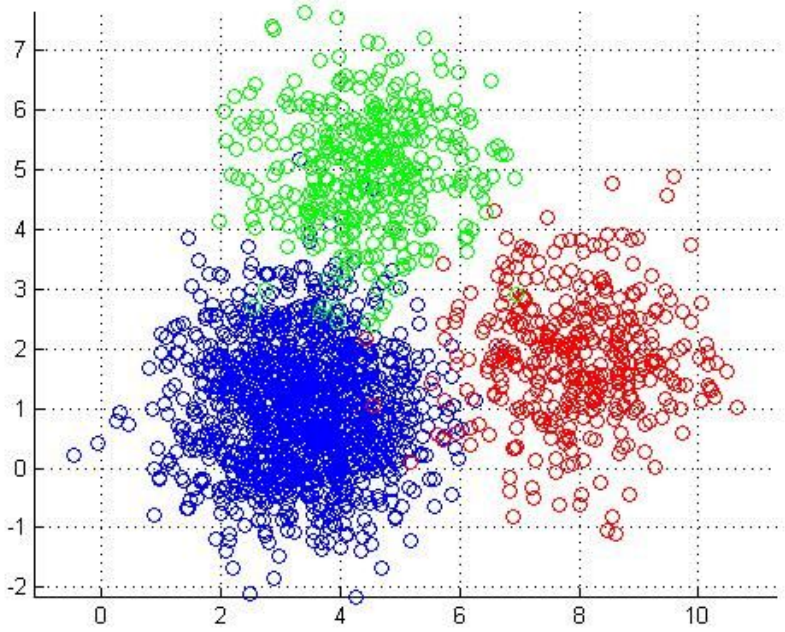
Brief Recap: K-Means Clustering

Given number of desired clusters k, find the optimal means for them, and classify points as such.

Unsupervised, Deterministic*, not a classifier

Complexity: O(nkdi) i=iterations

- What is a mean vector and its dimensions?



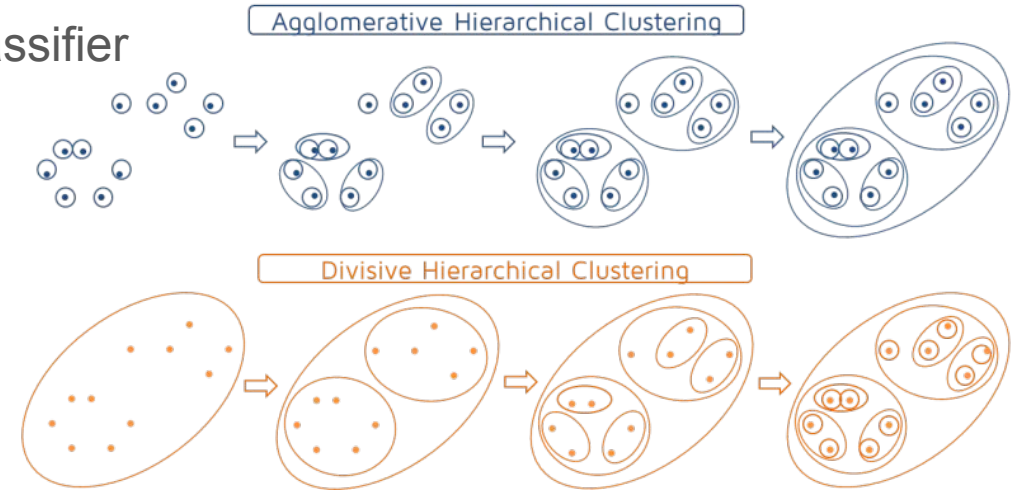
Brief Recap: Hierarchical Clustering

Bottom Up/Divisive: All points in one cluster

Agglomerative: Every point in its own cluster

Unsupervised, Deterministic*, not a classifier

Complexity: O(n^3)

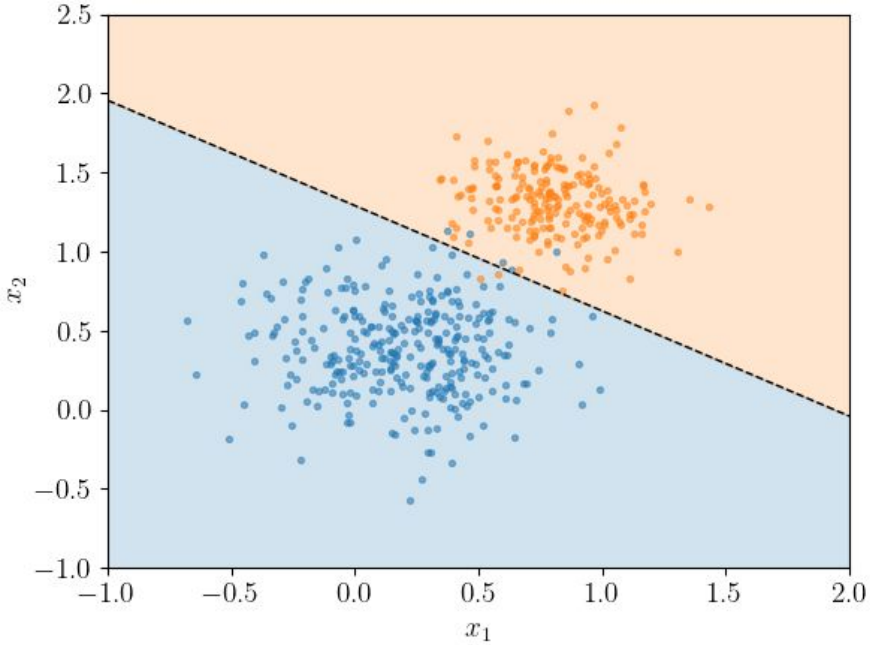


Brief Recap: Logistic Regression

Learn a linear decision boundary.

Supervised, Discriminative, probabilistic, linear

Time Complexity: O(nd^2)



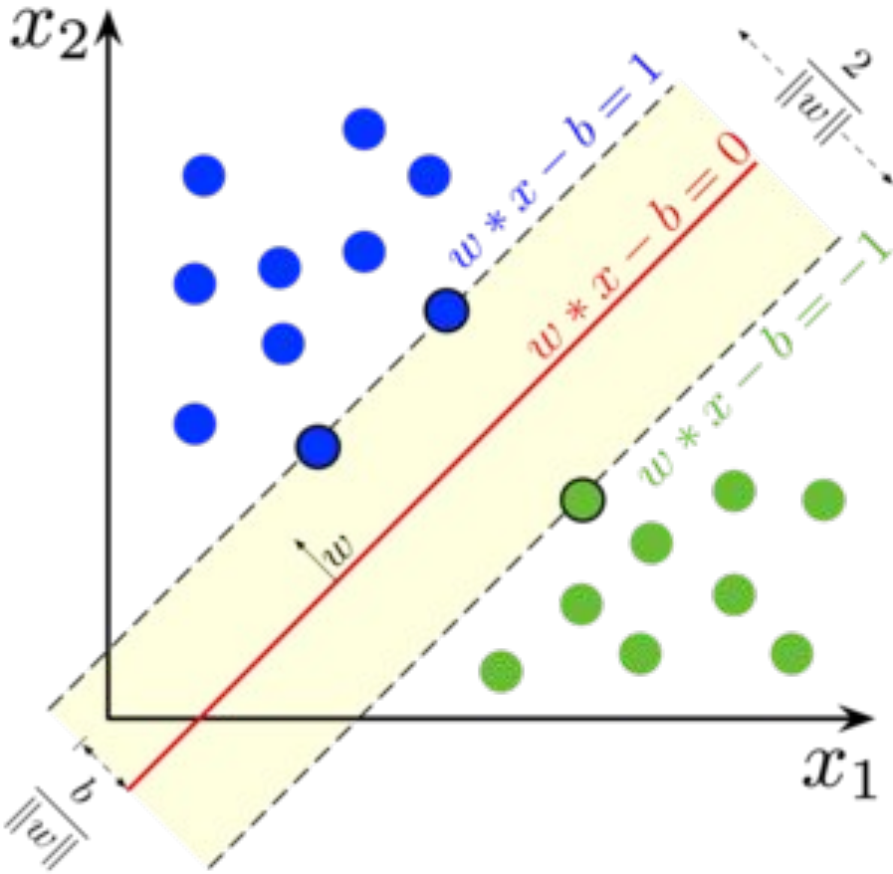
SVM Overview

Learns maximal margin between classes

Supervised, Discriminative, deterministic*, linear*

Training: O(n^3)

Prediction: O(nd)



SVM Primal and Dual Variables

Primal problem solves SVM using all the features.

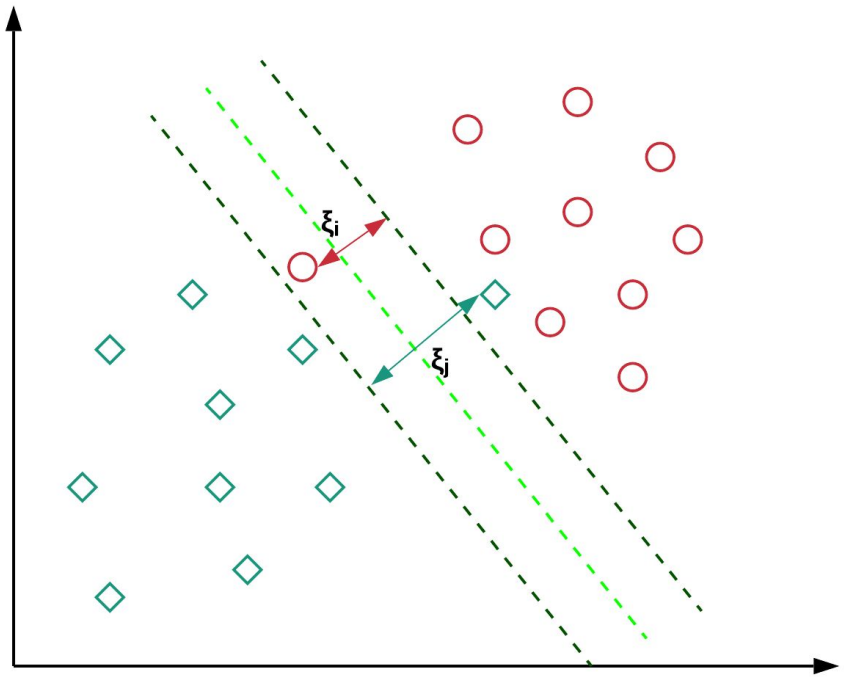
Dual problem uses Lagrangian multipliers to change problem into one that depends on dot products of the feature vectors, enabling kernel trick.

- How to convert between primal and dual?
- How to identify support vectors (mathematically and conceptually)?

SVM Non-separable case

If the 2 classes are not linearly separable, error term needs to be introduced.

- How does this affect solving for support vectors?



SVM vs Logistic Regression

1. Which outputs probabilities?
2. Which is more accurate?
3. Consider an unbalanced case, 10000 class labels 0, and 10 as 1. What will happen in each?

Continuous vs Categorical Data

What methods handle categorical data naturally?

What methods handle continuous data naturally?

How to get probabilities with continuous data?

How to get numbers out of categorical data?

Practice Problems

(a) True or False: Naive Bayes uses a generative model for classification.

(b) True or False: The training error of 1-NN classifier is 0.

(c) True or False: Linear regression is an unsupervised learning problem; logistic regression is a supervised learning problem.

(d) True or False: Logistic regression cannot be trained with gradient descent algorithm.

(e) True or False: The test data classification accuracy of K-nearest neighbor classifier increases as K increases.

(f) True or False: In K-means clustering, a larger K is always more desirable than a smaller K.

(h) True or False: Logistic regression is different from SVM such that logistic regression is a generative model due to its ability to provide uncertainty of the classification result.

(j) True or False: The global minimum is guaranteed to be reached by using gradient descent.

(a) True or False: Naive Bayes uses a generative model for classification. **True**

(b) True or False: The training error of 1-NN classifier is 0. **True**. Each point is its own neighbor, so 1-NN classifier achieves perfect classification on training data.

(c) True or False: Linear regression is an unsupervised learning problem; logistic regression is a supervised learning problem. **False**

(d) True or False: Logistic regression cannot be trained with gradient descent algorithm. **False**

(e) True or False: The test data classification accuracy of K-nearest neighbor classifier increases as K increases. **False**

(f) True or False: In K-means clustering, a larger K is always more desirable than a smaller K. **False**

(h) True or False: Logistic regression is different from SVM such that logistic regression is a generative model due to its ability to provide uncertainty of the classification result. **False**

(j) True or False: The global minimum is guaranteed to be reached by using gradient descent. **False**

Assume you collected neuron expression data for 30,000 genes from 300 alzheimer's patients. Consider using K-means clustering algorithm for unsupervised learning. How many mean vectors are there and what is the dimension of each mean vector, if you want to cluster patients to find $K = 10$ clusters of patients for alzheimer's subtype discovery.

Assume you collected neuron expression data for 30,000 genes from 300 alzheimer's patients. Consider using K-means clustering algorithm for unsupervised learning. How many mean vectors are there and what is the dimension of each mean vector, if you want to cluster patients to find $K = 10$ clusters of patients for alzheimer's subtype discovery.

10 mean vectors

Dimension: 30,000 each

We want to predict cancer severity with 100 clinical variables (such as family history, BMI, age, etc.). We have 700 patients with cancer. Set up a linear regression model.

What are the dimensions of X?

What is the dimension of Y?

How to solve?

We want to predict cancer severity with 100 clinical variables (such as family history, BMI, age, etc.). We have 700 patients with cancer. Set up a linear regression model.

What are the dimensions of X? **700 rows x 100 columns**

What is the dimension of Y? **700 rows (patients) x 1 column**

How to solve? **B = (XTX)^-1 XTY**